

Model Capacity, LASSO and Cross Validation

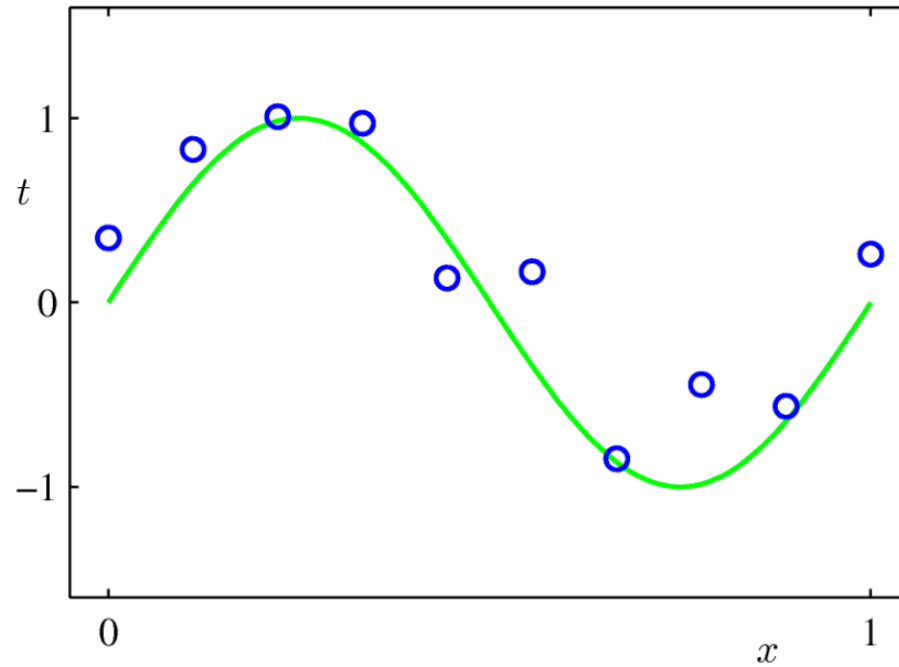
Capacity of a model

Model capacity is ability to fit variety of functions

- Model with Low capacity struggles to fit training set
- A High capacity model can overfit by memorizing properties of training set not useful on test set
 - When model has higher capacity, it overfits
- One way to control capacity of a learning algorithm is by choosing the hypothesis space
 - set of functions that the learning algorithm is allowed to select as being the solution
 - E.g., the linear regression algorithm has the set of all linear functions of its input as the hypothesis space
 - We can generalize to include polynomials in its hypothesis space which increases model capacity

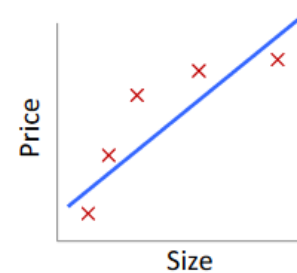
Capacity of Polynomial Curve Fits

$$y = \theta_0 x^0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_p x^p = \sum_{j=0}^p \theta_j x^j$$

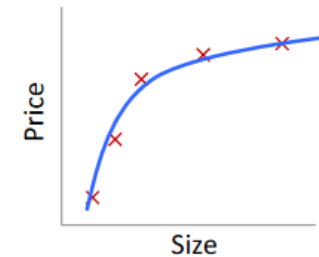


Appropriate Capacity

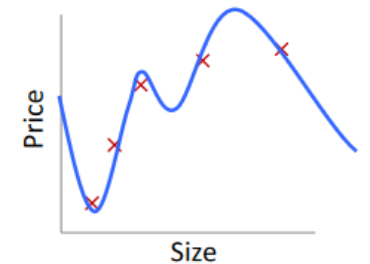
- Machine Learning algorithms will perform well when their capacity is appropriate for the true complexity of the task that they need to perform and the amount of training data they are provided with
- Models with insufficient capacity are unable to solve complex tasks
- Models with high capacity can solve complex tasks, but when their capacity is higher than needed to solve the present task, they may overfit



$\theta_0 + \theta_1 x$
Underfitting
(high bias)



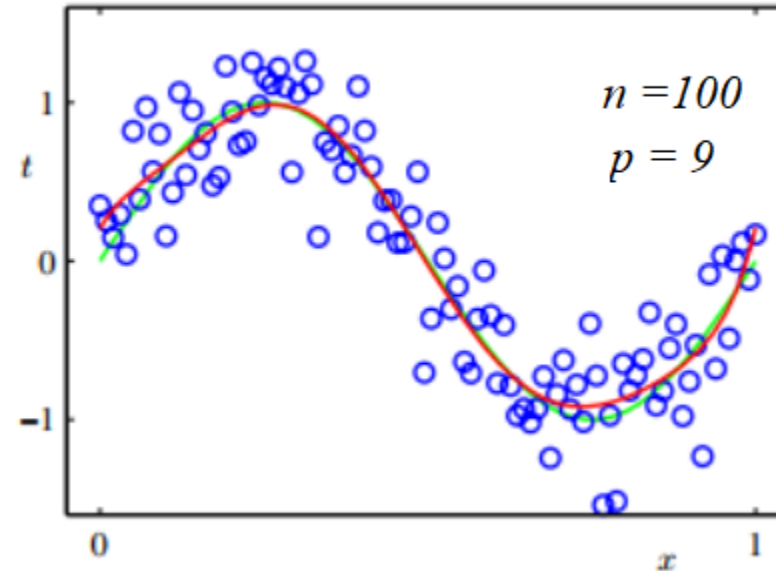
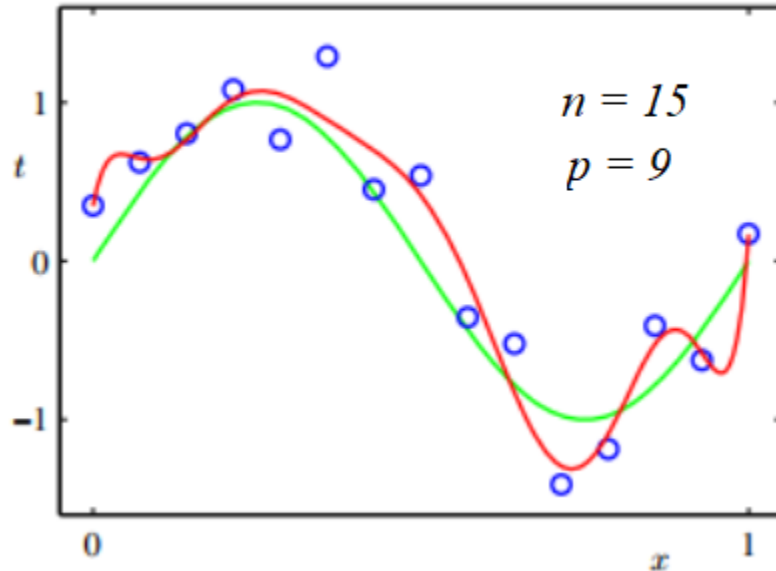
$\theta_0 + \theta_1 x + \theta_2 x^2$
Correct fit



$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$
Overfitting
(high variance)

Reducing Overfitting Issue

- Reduce the complexity or model capacity
- Regularization
- Increase the amount of data (5 or 10 times the number of parameters in the model)

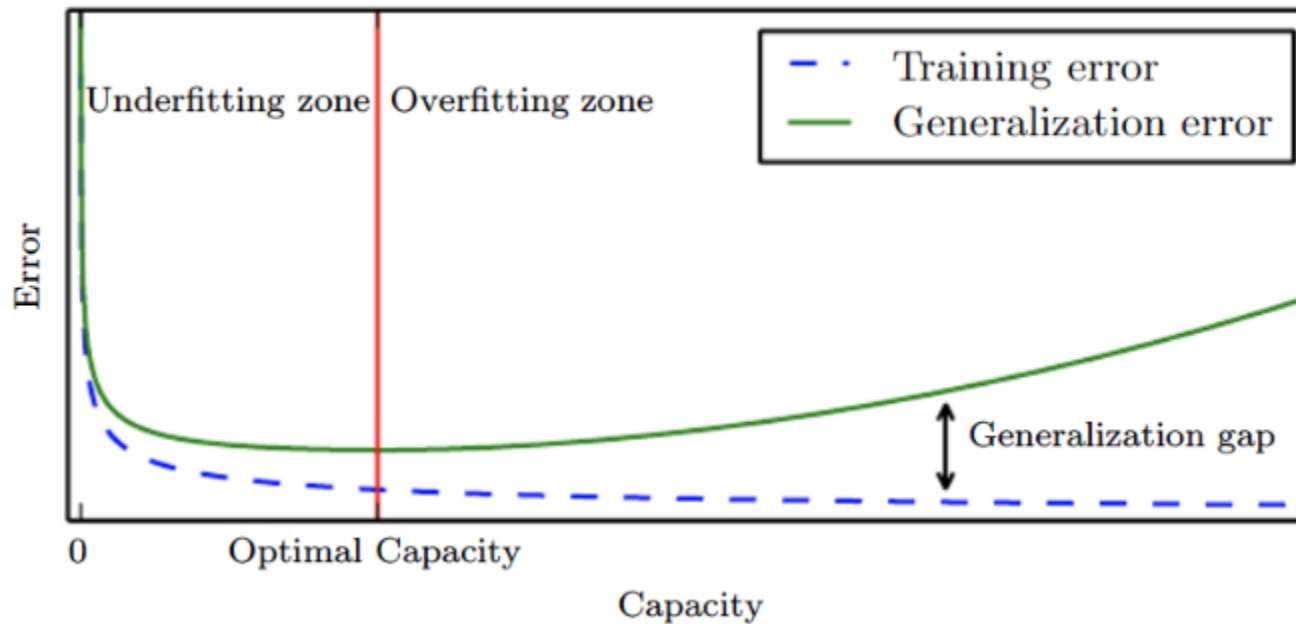


The Effective Number of Parameters

- Stack the outcomes $\mathbf{y} = [y^{(1)}, \dots, y^{(n)}]^T$ and predictions $\hat{\mathbf{y}}$.
- For a linear fitting method $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$
- $\mathbf{S}$ is an $n \times n$ matrix depends on the input vector $x^{(i)}$
- The number of effective parameters is defined as $d(\mathbf{S}) = \text{trace}(\mathbf{S})$

Typical Generalization Error

- Relationship between capacity and error
- Typically generalization error has a U-shaped curve



Least Absolute Shrinkage and Selection Operator (LASSO)

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \quad \|\vec{y} - \vec{X}\vec{\theta}\|_2^2 + \lambda \|\vec{\theta}\|_1$$

Special case: $\vec{X}^T \vec{X} = \mathbf{I}$.

$$J(\theta) = \vec{y}^T \vec{y} + \vec{\theta}^T \vec{X}^T \vec{X} \vec{\theta} - 2 \vec{y}^T \vec{X} \vec{\theta} + \lambda \|\vec{\theta}\|_1,$$

$$= \vec{y}^T \vec{y} + \vec{\theta}^T \vec{\theta} - 2 \vec{y}^T \vec{X} \vec{\theta} + \lambda \|\vec{\theta}\|_1,$$

$$J(\theta) = \vec{y}^T \vec{y} + \sum_{j=1}^d (\theta_j^2 - 2 \vec{y}^T \vec{x}_j \theta_j + \lambda |\theta_j|)$$

$\vec{x}_j = j$ th column of $\vec{X}$

$$J(\theta) = y^T y + \sum_{j=1}^d (\theta_j^2 - 2y^T x_j \theta_j + \lambda |\theta_j|)$$

For $\theta_j > 0$:- $2\theta_j - 2y^T x_j + \lambda = 0$

$$\hat{\theta}_j = y^T x_j - \frac{\lambda}{2}$$

For $\theta_j < 0$:- $2\theta_j - 2y^T x_j - \lambda = 0$

$$\hat{\theta}_j = \underline{y^T x_j + \frac{\lambda}{2}}$$

For $\theta_j = 0 \rightarrow \hat{\theta}_j = 0$

Ordinary Least Square soln $\hat{\theta}_j^0 = y^T x_j$

$$\hat{\theta}_j = \begin{cases} \hat{\theta}_j^0 + \frac{\lambda}{2} \\ \hat{\theta}_j^0 - \frac{\lambda}{2} \\ 0 \end{cases}$$

if $\hat{\theta}_j < 0$

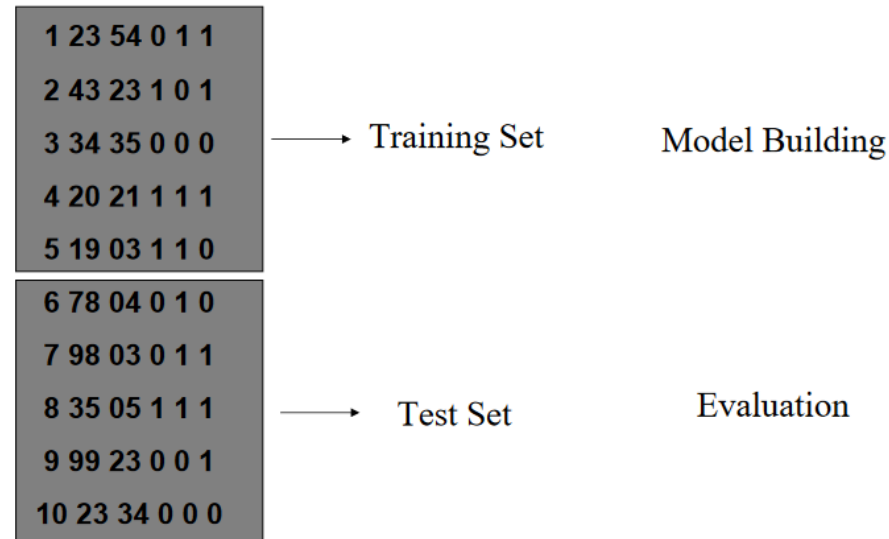
if $\hat{\theta}_j > 0$

else

Soft-Thresholding

Training and Testing Sets

- Training set is used to build the model
- Test set left aside for evaluation purposes
- Ideal: different data set to test if model generalizes to other settings
- If data are abundant, then there is no need to “recycle” cases



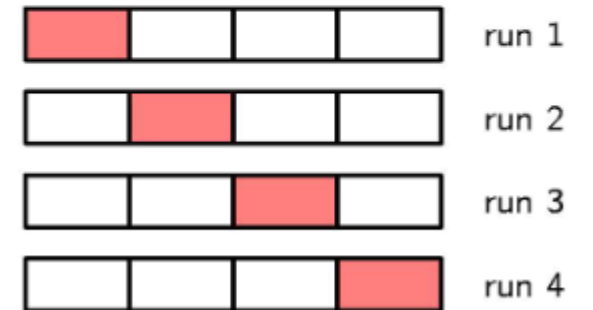
Cross-Validation

- The simplest and most widely used method for estimating prediction error is cross-validation.
- When data set is too small, dividing into a fixed training set and fixed testing set is problematic
 - If it results in a small test set
 - Small test set implies statistical uncertainty around the estimated average test error
 - Cannot claim algorithm A works better than algorithm B for a given task
- k - fold cross-validation
 - Partition the data into k non-overlapping subsets
 - On trial i , i th subset of data is used as the test set
 - Rest of the data is used as the training set

k - fold Cross-Validation

- Supply of data is limited
- All available data is partitioned into k groups (folds)
- $k - 1$ groups are used to train and evaluated on remaining group
- Repeat for all k choices of held-out group
- Performance scores from k runs are averaged

$k=4$



k - fold Cross-Validation

- $\kappa: \{1, 2, \dots, n\} \mapsto \{1, 2, \dots, k\}$ -- An indexing function that indicates the partition to which observation i is allocated by the randomization.
- The fitted function is $\hat{f}^{-k}(x)$ be the fitted function, computed with the k th part of the data removed.
- The cross-validation estimate of the prediction error is

$$CV = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}^{-\kappa(i)}(x_i))$$

- Typical choices of K are 5 or 10.
- $k = n$ is known as *leave-one-out* cross-validation.